OPERATION AND MAINTENANCE MANUAL – PART 1

EXCITATION SYSTEM DESCRIPTION

DOCUMENT NO.: VIN/90/PCS/400018/EX/OMM001/A

Plant Name : VINH-SON HYDRO POWER STATION
Location : VIETNAM
Plant Type : STATIC EXCITATION SYSTEM
Plant Rating : 40MVA

A	Revised after Commissioning	2011-10-20	HL Cheng	Davor Tomerlin	Davor Tomerlin
-	Initial Issue	2011-04-20	HL Cheng	Davor Tomerlin	Davor Tomerlin
Rev	**Description**	**Date**	**Prepared**	**Checked**	**Approved**

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REVISION HISTORY

REV	CREATED BY	CHECKED BY	APPROVED BY	DATE	DESCRIPTION	STAT.
-	HL Cheng	Davor Tomerlin	Davor Tomerlin	2011-04-20	Initial issue	PREL
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Chapters		Remarks	
		Yes	No

Section 1 DESCRIPTION OF THE EXCITATION SYSTEM

1.1 General description

Generally, the excitation system is applied primarily as a voltage regulator for the synchronous generator. The primary function of the excitation system is to provide variable field current to the rotor winding of the generator and ultimately inducing ac voltage in stator winding. In controlling the field current, the stator voltage can be regulated.

The proposed excitation system is of static type and is based on a digital automatic voltage regulator (AVR), which controls the firing angle of the thyristor bridges so as to supply the generator with variable field current.

The proposed equipment configuration is of redundant channel configuration and configurable according to the requirements of the control system. The system also consists of two Thyristor rectifier bridges. A failure of the active Thyristor Bridge causes the system to change over to the stand-by Thyristor Bridge.

The excitation system hardware is housed in a sturdy and compact cubicle. The regulator design avoids any wiring faults and ensures good EMC performance.

1.2 Working conditions

Minimum working temperature	-5	°C
Maximum working temperature	+45	°C
Relative humidity without condensation	93	%
Max. altitude for installation	1000	m

1.3 Equipment main characteristics

Excitation transformer

Number of phases	3	
Rated power	456	kVA
Operating frequency	50	Hz
Primary voltage	13.8	kV
Secondary voltage	…220.	V
Vector symbol	yd11	
Mass	kg	
Protection index	IP	…

Electrical Braking transformer

Number of phases	3	
Rated power	37	kVA
Operating frequency	50	Hz
Primary voltage	380	V
Secondary voltage	38	V
Vector symbol	Dy5	
Mass	205	kg
Protection index	IP	00

Excitation cubicle

Regulation structure	2/2	
Number of rectifier bridges	2	
Rated field current	1179	A
No load field current	…599.	A
Maximum field current during normal excitation	1297	A
Field current during ceiling excitation (10 sec.)	2358	A
Rated excitation voltage	86	V
Maximum excitation voltage	250	V
Protection index equipment	IP	31

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1.4 Voltage regulator main characteristics

REGULATOR TYPE	**ALSPA AVR**			
NUMBER OF CHANNELS	**1**			
REGULATOR STRUCTURE	☐ AVR	☑ AVR/MCR	☐ AVR/AVR	☐ Surveillance conduction thyristor

DIGITAL REGULATOR FUNCTIONS	**COMMENTS**
☑ STATISME ACTIF et REACTIF	
☑ OVER EXCITATION LIMITATION	
☑ UNDER EXCITATION LIMITATION	
☑ REACTIV POWER REGULATION	
☑ POWER FACTOR REGULATION	
☑ STATOR CURRENT LIMITATION	
☑ ROTOR TEMP. MONITORING	
☑ FLASHING SEQUENCE	
☐ BOOSTING SEQUENCE	
☐ MAX. EXCITATION CURRENT SEQUENCE	
☐ STATOR VOLTAGE LIMITATION Automatic	
☐ STATOR VOLTAGE LIMITATION manual	
☐ OVER FLUXING LIMITATION	
☐ GRID VOLTAGE REGULATION	
☑ PSS Standard	
☐ PSS 1A	
☑ PSS 2A	
☑ VOLTAGE SET POINT REMOTE CONTROL	
☐ REACTIVE POWER SET POINT REMOTE	
☐ POWER FACTOR SET POINT REMOTE CONTROL	
☐ EXCITATION CURRENT SET POINT REMOTE CONTROL	
☐ WORLDFIP LINK TO DCS	
☐ MODBUS SERIAL LINK TO DCS	
☐ OPTICAL FIBER	
Others	
4-20 ma Outputs :	**Observations**
☑ EXCITATION CURRENT	
☑ EXCITATION VOLTAGE	
☐ STATOR VOLTAGE	
☐ ROTOR TEMPERATURE	
Others	
0-10V Outputs :	**Observations**
☐ ROTOR TEMPERATURE	
☐ CONTROL SIGNAL DIFFERENTIAL (Auto/Manu)	
Others	

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1.5 Regulation system

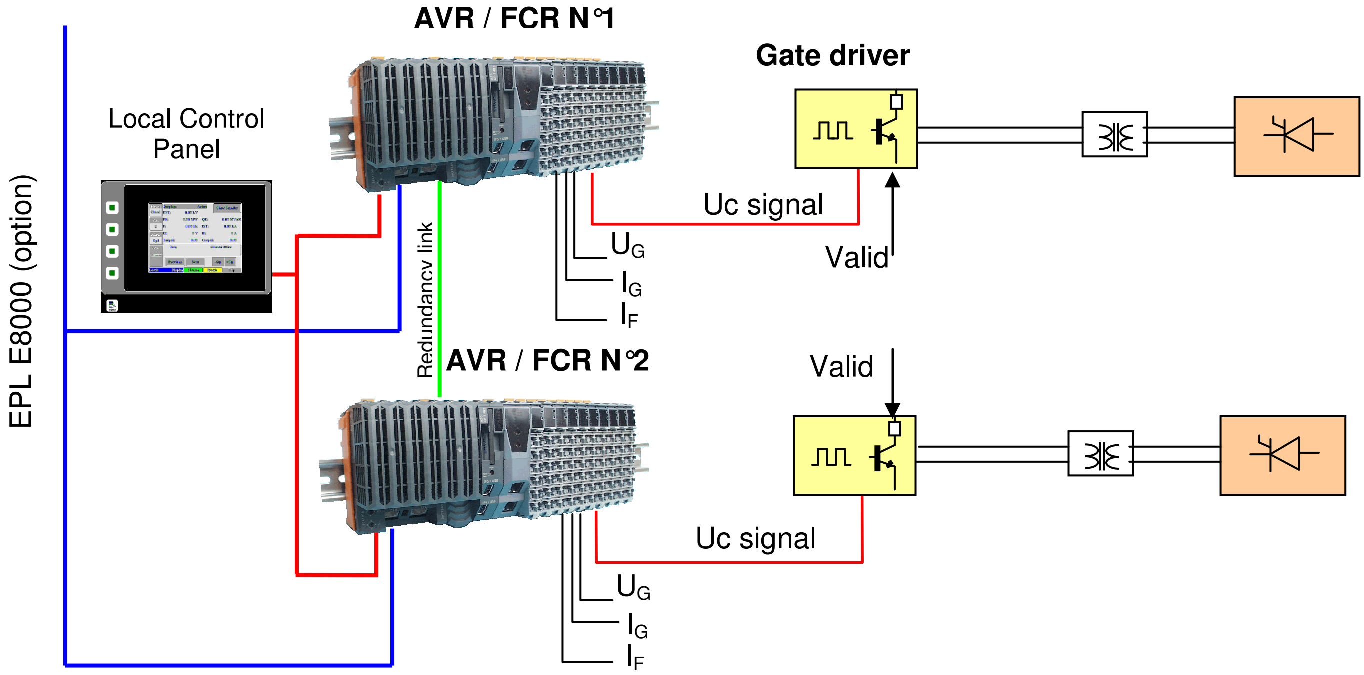
1.5.1 Regulation structure

The regulator configurations is:

- Fully redundant AVR/FCR configuration with two regulators,

1.5.1.1 Fully redundant AVR/FCR configuration with two regulators

The system consists of two identical digital voltage regulators, Regulator 1 and Regulator 2, both with an integrated FCR (Field Current Regulator). Each regulator has its own power supply and its own digital thyristor-firing module. A failure of the active regulator causes the system to change over to the stand-by regulator. A tracking function allows a smooth transfer from one regulator to the other.



Such a structure with a twin automatic regulation and integrated FCR is fully redundant and ensures best availability of the power generating system.

With the redundant thyristor rectifiers, each regulator can control either one of the rectifiers (for independent redundancy between the regulators and the rectifiers).

1.5.2 Regulator functions

The main functions of the system are:

Power supply to the control system: The regulator and its auxiliaries need a safe and reliable power source to avoid any risk of generator shutdown due to faults in auxiliary devices.

Power supply to the excitation system: The excitation power can be provided by different sources, e.g. by an excitation transformer.

Flashing excitation: An external power supply is needed to start the generator. This power source is provided by power station's 125V DC battery power supply.

Rotor overvoltage protection: There are two type of rotor overvoltage protections provided within the excitation system as follow: -

1) Overvoltage Protection (OVP): It is generally performed by a crowbar system through break-over diode. This protection is provided by linear discharge resistors.

2) Discharge of the field windings: Prior to field breaker opening, the rotor windings have to be discharged to prevent overvoltage across the main contacts of the field breaker. This is carried out by means of linear discharge resistor together with the closing of field breaker's rupturing pole.

Thyristor rectifier: The thyristor rectifier provides variable field current.

Measuring system: The regulator requires the acquisition of some generator values, such as stator voltage, stator current and rotor current.

Automatic voltage regulator: For improved stability, the automatic voltage regulator controls the generator voltage by calculating the firing angle of the thyristors in the rectifier bridge.

Interface system: In many applications, the regulator is linked to a existing conventional control panel in order to exchange e.g. variables, status and fault data. The regulator also needs to be configured, or some parameters may have to be adapted to the generator. For this purpose the excitation system is fitted with communication links to connect the CPU with the laptop with Controcad software.

For further details about the Interface system, refer to Part 2.

These functions will be described in the following sections.

1.5.3 Limitations

1.5.3.1 V/Hz limitation

The limitation function consists in replacing the voltage set point by the output of a minimum or maximum U/F ratio selector circuit.

The maximum and minimum U/F values are pre-adjusted and the maximum frequency is limited by a pre-adjusted value.

1.5.3.2 Under-excitation limitation

This function is designed to work out a reactive power absorption limiting signal in order to keep the generator within its stable operating range.

The function integrates two independent types of limitation, allowing a good adjustment of the limitation according to the reactive power capability curve of the machine:

- The first loop is acting on the reactive power signal preventing its value to decrease below a minimum negative value. The reactive power limit is designed thanks to an adjustable three segments line in generator capability curves.
- The second loop is acting on the field current signal keeping it above a minimum value.

When the limitation loop input signal becomes lower than its permitted minimum value, the corresponding Integral-Proportional regulator generates a voltage output signal. This output signal is substituted to the actual stator voltage set-point in order to maintain either the reactive power or the field current equal to its minimum allowed value. Then the requested stator voltage set-point is no more realised.

This structure allows to keep the PSS in operation at every time.

1.5.3.3 Stator current limitation

This function is designed to keep the stator current within its limits and thus to avoid overheating of the stator. As an option, this limitation might be temperature dependent (two slopes to approximate generator temperature capability curve).

The limitation provides an over excitation signal when QE < 0 and a de-excitation signal when QE > 0. The stator current limitation reference presents an “Inverse time” behaviour. That means that the set point decreases according to the overheating of the stator.

When the automatic mode is not operating, the output is forced to zero

When QE approaches zero limitation is disabled (dead band).

Inside the dead band the limitation function is no longer operating. The only way to limit the stator current is then to send a signal in order to limit the power of the turbine.

1.5.3.4 Power system stabiliser

The aim of this system is to smooth rotor oscillations by elaborating a signal which acts against the angle variation (= load angle + grid angle).

Two kinds of PSS functions are available:

- A single input PSS, so-called PSS1A in reference with the IEEE Std 421.5: it input signal is the active power.
- A dual input PSS, so-called PSS2B in reference with the IEEE Std 421.5: its input signal is the integral of the accelerating power. It is calculated from the speed and an integral of electrical power, using the mechanical model of the shaft line, linked to its inertia.

The main components of the PSS function are:

- The high-pass filters (so-called washout filters) are classically used on the PSS input signals. They eliminates their DC component.
- The lead-lag filters introduce the necessary phase compensation in the frequency range in which the PSS shall have a positive damping effect.
- The PSS gain.
- The PSS output limitation that avoids too strong action of the PSS in case of large disturbances.

1.5.3.5 Overexcitation limitation

The over-excitation limitation aims to obtain a maximum operation of the generator according to the rotor heating. As an option, this limitation might be temperature dependent (two slopes to approximate generator temperature capability curve).

The function limits, in AUTO mode, the rotor current in order to avoid generator overheating. The limitation includes two different functions:

- The thermal limitation allows excitation current to be greater than the permanent thermal excitation current for a defined duration linked to the generator overheating limits. So the function acts during grid faults where the machine is solicited for contribution to grid stability keeping, and ensures that the machine will not suffer from overheating damages.
- The ceiling limitation that prevents the excitation current to increase above the ceiling current of the machine.

Thermal limitation:

The thermal limitation is a proportional and integral function. It controls the generator field current.

Two possibilities are available for the calculation of the limitation reference. But when the limitation is not active, its set-point is normally adjusted to the ceiling reference value allowing high field current.

Instantaneous limitation:

An instantaneous logic calculation determines the reference of the limitation:

When the field current increases above the permanent reference value, this over-current is allowed for a fixed time delay. This time delay is specified by the generator constructor, and corresponds to the allowed time for ceiling current application. After this time delay, if the field current is still above the permanent reference value, then the limitation set point is reduced down to the permanent reference value.

The return to the initial value of the limitation reference is again possible instantaneously when the field current stays under a lower value (proportional to the permanent reference value) for a specified duration.

Inverse time limitation (Option):

An inverse time calculation determines the reference of the limitation:

In this case, the limitation set-point decreases from its maximal reference value to the permanent reference value according to the overheating of the rotor. The limitation reference takes into account the level of the actual excitation current and the decrease is slaved to the gap between the excitation current value and the permanent reference value. The return to the ceiling reference value is also gradual.

Ceiling limitation:

The set point of the ceiling limitation is adjusted to the permitted ceiling current value and the field current controlled by a PID function.

The output signal over-rides the AVR output.

1.5.3.6 Droop

Two kinds of droop function are available:

- an active and reactive compensation function,
- a frequency droop function.

The active and reactive compensation function allows to operate several generators either connected directly in parallel (negative compensation gain), or to compensate the line voltage drop (positive compensation gain) through the Main Transformer.

This function generates a corrective term added to the voltage set point, proportional to the actual active and reactive stator current. The correction gains can be set from -20% to +10%.

The frequency droop function allows to increase the generating unit stability when operating in a local area isolated network.

With this function acting on the voltage set point, the stator voltage is increased as a function of increasing frequency, within a limited range. The correction gain can be set from 0% to 5%.

1.5.3.7 Stator voltage limitation in manual mode (option):

As an option a stator voltage limitation function may be available in manual mode.

1.6 Excitation digital protection

Excitation transformer and excitation equipment are protected by protection relays.

1.6.1 MX3IPG2A

64 : Rotor earth fault protection

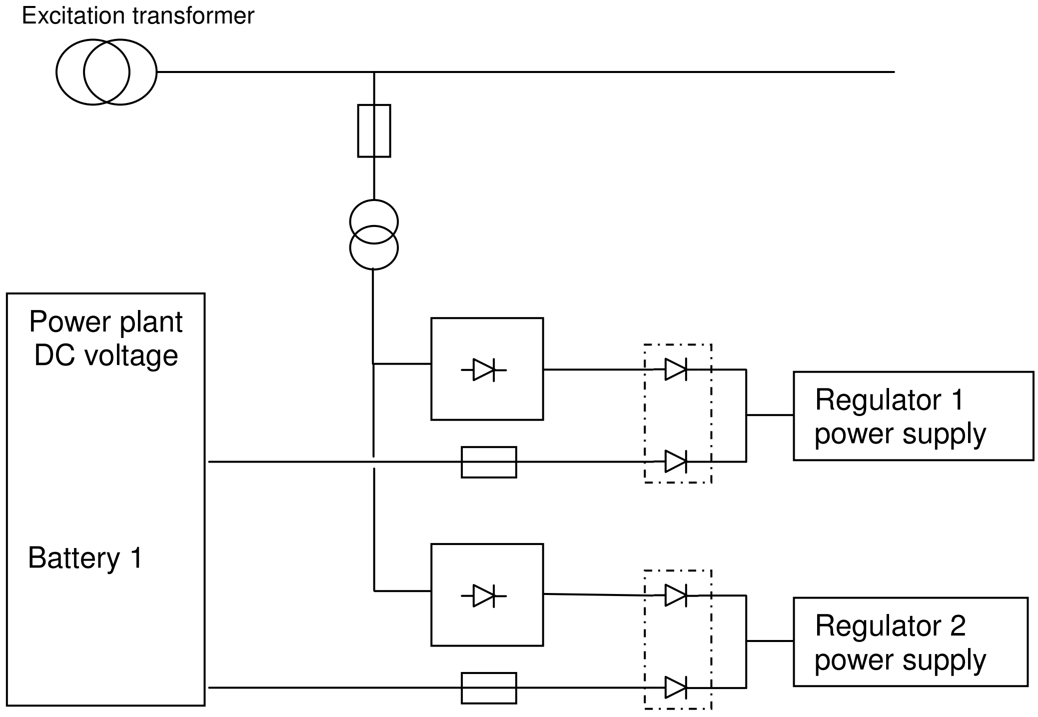
The relay monitors the insulation of the rotor winding. A single fault will not effect generator operation but needs to be eliminated before occurrence of a second insulation fault, which may cause severe generator damage.

76: Excitation DC Overcurrent Protection

The relay monitors the Excitation current by taking in DC Voltage Measurement via Shunt. The first stage fault gives an alarm on LCP. The second stage fault gives a tripping order immediately when the excitation current exceeds a reference threshold.

1.7 Power supply to the control system

The control system is fed by two different power sources. One is the plant battery, the other is the excitation transformer or the braking transformer depend on the mode of operation, which supplies rectified AC. Both sources are connected in parallel for the purpose of redundancy.

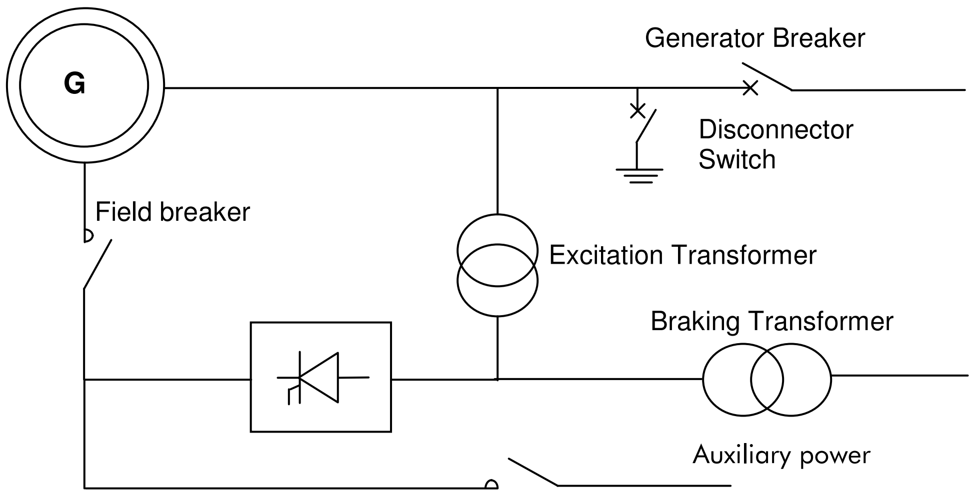


The control system power supplies are in a mutual backup configuration. When either the AC or the DC source fails, the other one is able to supply the whole system.

1.8 Power supply to the excitation system

1.8.1 Excitation current supplied from shunt excitation transformer & electrical braking transformer

The power necessary for generator excitation is supplied by an excitation transformer connected to the generator output.



In normal operation, the excitation transformer is fed directly by the generator. The field current is provided by the secondary windings of the excitation transformer. The Thyristor Bridge controls the generator voltage by varying the average field current value. During the start-up sequence, the magnetic field of the generator is not strong enough to produce an output voltage. As long as no current is supplied by the excitation transformer, the rotor is powered by 125VDC from plant DC battery bank. This auxiliary excitation system is usually called "Flashing circuit".

Under normal stopping sequence, mechanical braking is used to stop the hydro-generator and to bring the turbine & rotor to a standstill condition. This conventional braking function is based on principle of mechanical friction, which slows down the rotating mass of the generator.

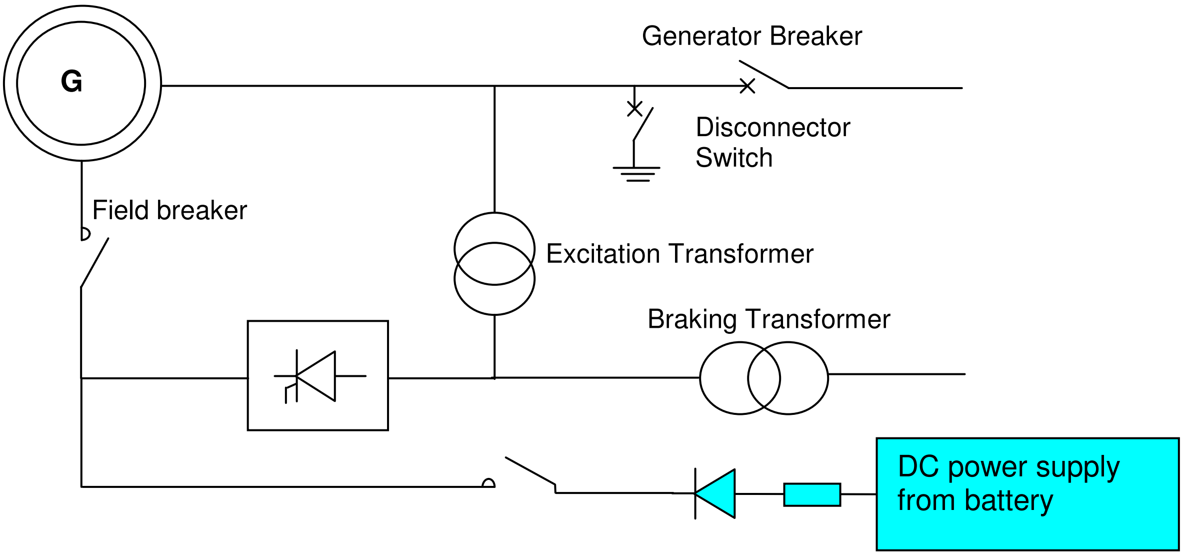
When electrical braking mode is selected, the braking sequence will be initiated after deloading and disconnecting the generator from the network, typically when the generator has reached 80%-90% of the rated speed. The braking sequence will start with the changeover sequence to changeover the supply from excitation transformer to braking transformer. The generator armature/stator is then short-circuited by closing the disconnector switch.

The excitation system will be then be activated under manual mode to excite the generator to an approximate constant value of the excitation current which induces a rated armature current. This braking sequence continues until the generator standstill.

This method considerably shortens the generator stopping time as compare to conventional mechanical braking. The electrical braking sequence will typically end when the generated speed goes below 1% of the rated speed. After the braking sequence is completed and when the generator is in standstill condition, all the devices/equipment will be resumed to their initial position and the generator will be ready for a new start.

1.9 Flashing excitation

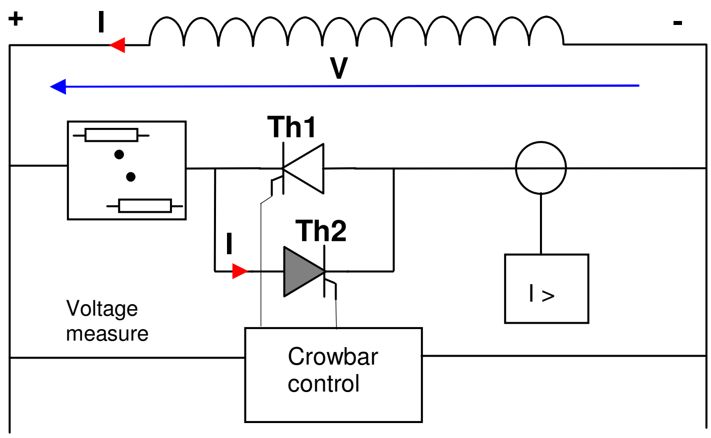
The flashing function is used with shunt type excitation systems. The flashing circuit is designed to supply the rotor field with current during the start-up sequence of the generator. This current is supplied by a DC battery. The source is typically designed to supply up to 15% of the rated no-load field current of the generator. (This value can be adapted according to the generator type). The flashing circuit is opened when the generator voltage reaches 15% of its rated value.



1.10 Overvoltage protection by crowbar

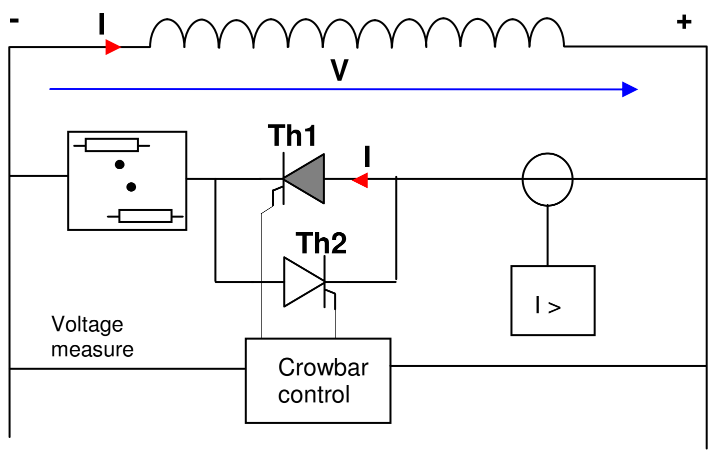
This function connects a set of resistors to the rotor windings, in order to rapidly reduce the generator excitation current in case of overvoltage caused by a short-circuit across the terminals of the generator. Since the overvoltage can be positive or negative, this protection is performed by means of two thyristors connected top to bottom. The voltage regulator includes no circuit for thyristor control; this is performed by a special electronic board without any external power supply. When the crowbar is used, the thyristor bridges of the excitation system are forced to inverter mode.

Protection against positive overvoltage



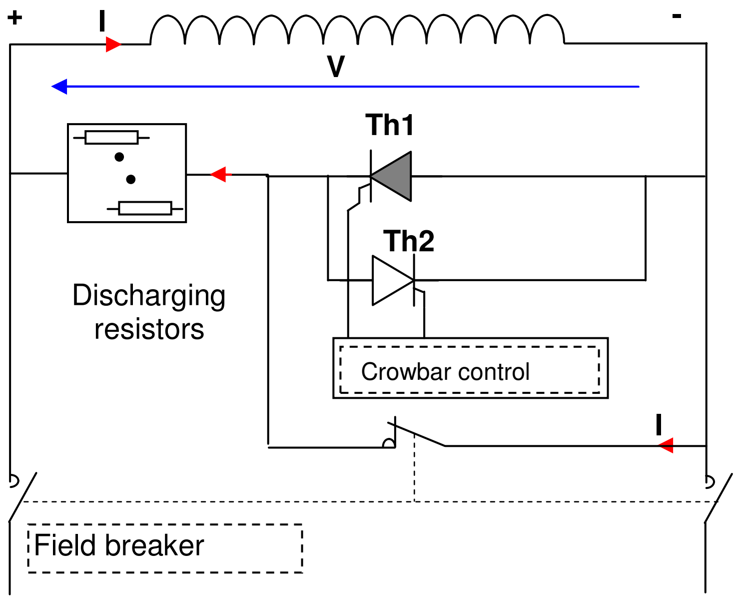
In case of a positive overvoltage, thyristor Th2 is forward biased and the overvoltage is attenuated through a resistor network. A current relay detects a current in the crowbar circuit and gives a tripping order to the excitation system.

Protection against negative overvoltage



In case of reverse power from the generator field winding, thyristor Th1 is forward biased and the energy of the winding is discharged through Th1 and the discharging resistors. A current relay detects a current in the crowbar circuit and gives a tripping order to the excitation system.

1.11 Discharge of the field windings



When the field breaker is opened, the current in the windings is discharged through a set of linear resistors. These resistors are connected by means of an auxiliary NC contact of the field breaker, which is closed 1 to 3 ms before the opening of the main contacts (NO). The rating of resistors depends on the energy to be discharged and their values are calculated such as to limit the peak voltage to a value less than the maximum rupturing voltage of the field breaker.

1.12 Thyristor rectifier

The thyristor rectifiers are designed to rectify and modulate the AC voltage from the excitation transformer and to control the field current of the generator. The thyristor rectifier designed for 3 AC phases are Graetz bridge equipped with quick-acting fuses (each with an auxiliary contact), an RC circuit for overvoltage limitation, a gate control circuit and a heat sink.

Cooling is of forced air type (REDEX 200/300 type rectifiers). The forced air cooling system includes an air duct with one fan fitted at the bottom of the frame. The cooling air flow monitoring is done by a differential pressure sensor

There are two rectifier bridges to serve the redundant configuration. In case of a rectifier failure, a monitoring system will automatically disable the bridge concerned.

The power rectifiers are designed to handle continuously 110% of the nominal excitation current, as well as the ceiling excitation current for a limited period.

The bridges are fed by the excitation transformer or electrical braking transformer, connected to terminals labelled L1, L2, L3 on the busbars of the bridge.

The DC output terminals are labelled L+ for positive polarity, and L- for negative polarity.

1.12.1 Forced air type cooling

Cooling is of forced air type (REDEX 200/300 type rectifiers). The forced air cooling system includes an air duct with one fan fitted on the top of the frame.

1.12.2 Over heating protection by heat sink temperature switch

The thyristor bridge is protected against overheating by a temperature switch. Temperature sensors are fitted on the top of the heat sink.

1.12.3 Over heating protection by temperature switch in the cooling duct

The thyristor bridge is protected against overheating by a thermostat. The thermostat sensor is located inside the cooling duct.

1.12.4 Cooling air flow monitoring

The cooling air flow monitoring is done by a differential pressure captor.

1.12.5 Three phases Thyristor Converter

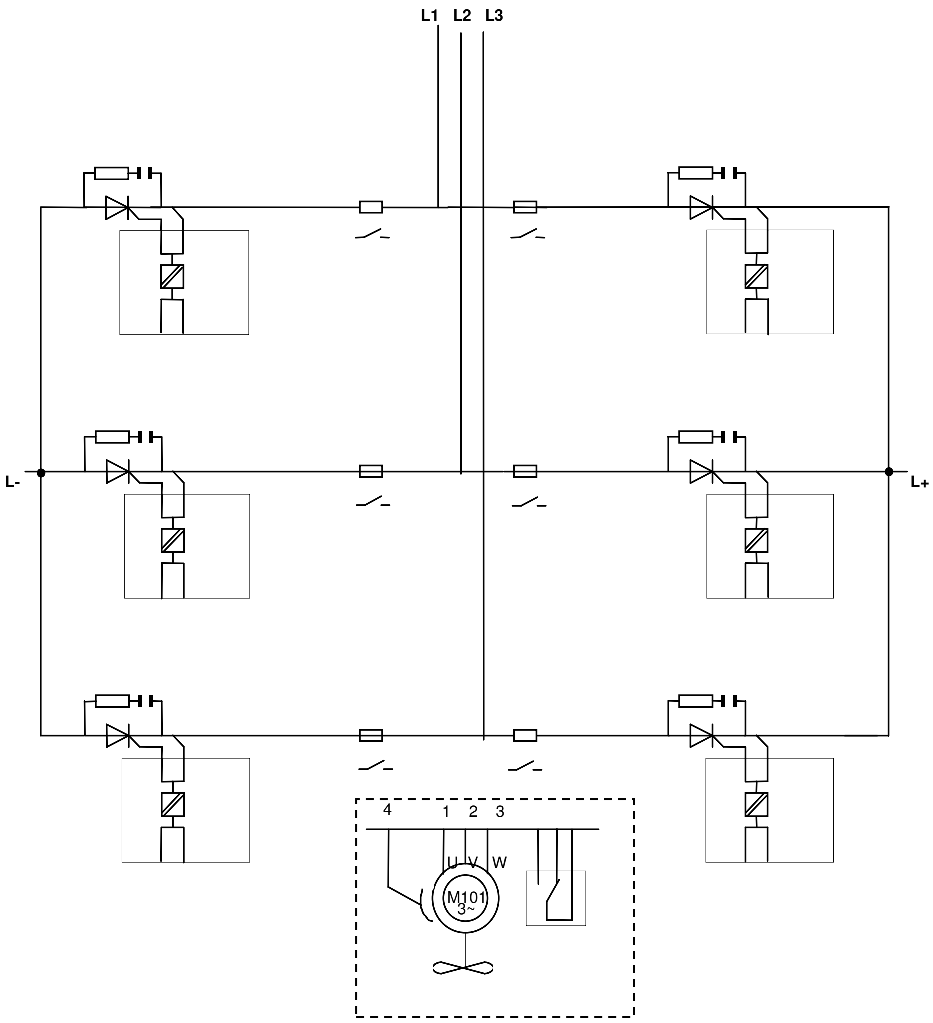
A full-wave three-phase thyristor bridge which is air force cooled by the fan is fitted within the cubicle.

SEMIKRON type thyristor bridge has the following characteristics:

- Six (6) thyristors, mounted on heat sinks, are designed for easy removal and replacement off line;
- The thyristors can withstand, at full operating temperature, a reverse voltage during the blocking periods of not less than 2,7 times the peak voltage;
- A current limiting high-speed fuse is provided to isolate the faulty branch and to protect other thyristors and their fuses from damage;
- Thyristor fuses blown monitoring;
- Over voltage snubber circuits per each thyristor are included to restrict the commutation over-voltages.

The Thyristor Bridge is equipped with temperature detector for overheating protection.

The airflow pressure is monitored by differential pressure device.



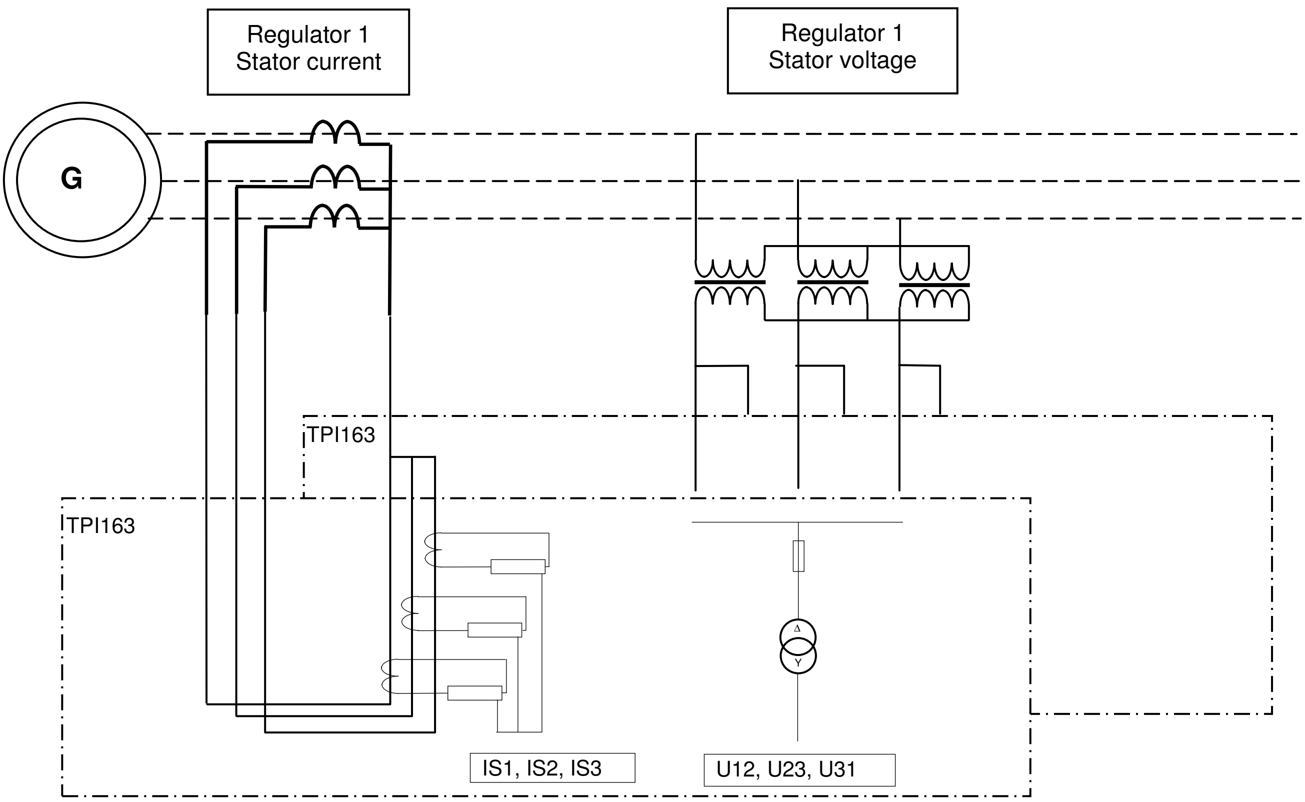
1.13 Measuring system

The regulation system needs information to process the control loops. The main values to be measured are:

- Generator stator voltage U12 – U23 – U31
- Generator stator current IS1 – IS2 – IS3
- Generator rotor current I_F

1.13.1 Arrangement for stator voltage and current measurements

The Stator current is measured by means of "**Current transformers**" whereas the "**Voltage transformers**" are used for Stator voltage measurements. Each regulator has its own voltage and current measurement system via theTPI163 input module. Only such a configuration can ensure full redundancy.



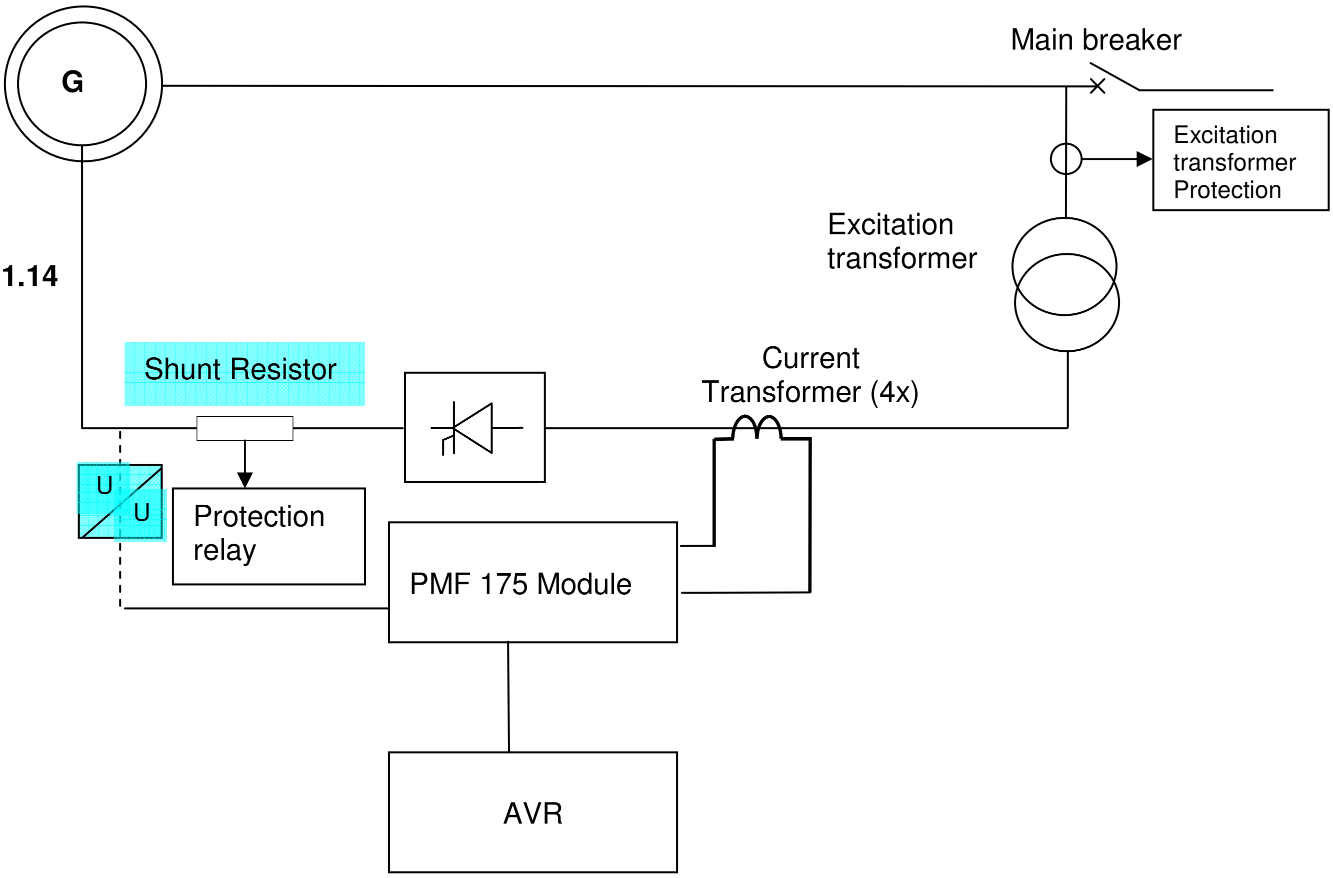
1.13.2 Rotor current measurement configuration

1.13.2.1 Measurement by “Shunt Resistor”

The rotor current is measured through a shunt resistor for DC voltage current measurement input to protection relay.

1.13.2.2 Measurement by Current Transformer

The current upstream of the thyristor bridges are measured by means of four current transformers, which further input into AVR through PMF175 measurement module for control purpose.



1.14

Voltage regulator

The Automatic Voltage Regulator (AVR) controls the generator output voltage so as to improve the generator's operational stability and keep it within its working limits. The AVR is a digital type and the regulation loops are processed by a CPU board. Several specific boards are connected to the main CPU board to perform electrical Input / Output functions (both logic and analogue).

The input/output boards are linked to terminal plates (for user connections) by means of ribbon cables connected in front of the I/O boards.

LEDs in front of the electronics boards indicate the status of the board itself and/or the corresponding I/Os.

The main available electronic boards, according to the regulator type, are the following:

Name of board	Description of the electronic board
X20CP1485	Central Processing Unit
X20IF1082	Ethernet powerlink interface module
X20AO4632	Analog output module, 4 outputs, +/- 10 V, 0.. 20 mA
X20DI9371	Digital input module, 12 inputs, 24VDC
X20DO8332	Digital output module, 12 outputs, 24VDC
X20AI4632.1	Analog input module, 4 inputs, +/- 11 V, 0.. 22 mA
X20TB12	Terminal block 12 pins (for each module)
X20BC0087	Bus controller field bus interface TCP/IP
X20PS9400	Supply module for bus controller
X20BB80	Bus base for bus controller and supply module
X20AT4222	4 inputs for resistance temperature measurement
PMF175	Measurement filtering module
TPI163	Interface for scaling the stator VTs & CTs
LCP	Local control panel
FL SWITCH LM8TX	Ethernet Switch
HUB 8TX-ZF	Ethernet Hub
TTM211	Thyristor gate controller board

1.14.1 Regulator rack arrangement

	IF1082		TB12	TB12	TB12	TB12	TB12	TB12
CP1485		dummy	AO	DI	DI	DO	AI	AI
		BM11	BM11	BM11	BM11	BM11	BM11	BM11

1.14.2 Distributed IO arrangement

	TB12		TB12	TB12	TB12	TB12	TB12	TB12	TB12	TB12	TB12
BC0087	PS9400	dummy	AI	AT	AO	DI	DI	DI	DI	DO	DO
BB80		BM11	BM11	BM11	BM11	BM11	BM11	BM11	BM11	BM11	BM11

1.14.3 Thyristor Bridge 1 Remote IO

	TB12		TB12	TB12
BC0087	PS9400	dummy	DI	DO
BB80		BM11	BM11	BM11

1.14.4 Thyristor Bridge 2 Remote IO

	TB12		TB12	TB12
BC0087	PS9400	dummy	DI	DO
BB80		BM11	BM11	BM11

1.14.5 Regulator architecture arrangement